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Review

Transforming Farming: A Review of AI-Powered UAV Technologies in Precision Agriculture

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Abstract: The integration of unmanned aerial vehicles (UAVs) with artificial intelligence (AI) and machine learning (ML) has fundamentally transformed precision agriculture by enhancing efficiency, sustainability, and data-driven decision making. In this paper, we present a comprehensive overview of the integration of multispectral, hyperspectral, and thermal sensors mounted on drones with AI-driven algorithms to transform modern farms. Such technologies support crop health monitoring in real time, resource management, and automated decision making, thus improving productivity with considerably reduced resource consumption. However, limitations include high costs of operation, limited UAV battery life, and the need for highly trained operators. The novelty of this study lies in the thorough analysis and comparison of all UAV-AI integration research, along with an overview of existing related works and an analysis of the gaps. Furthermore, practical solutions to technological challenges are summarized to provide insights into precision agriculture. This paper also discusses the barriers to UAV adoption and suggests practical solutions to overcome existing limitations. Finally, this paper outlines future research directions, which will discuss advances in sensor technology, energy-efficient AI models, and how these aspects influence ethical considerations regarding the use of UAVs in agricultural research.

Keywords: artificial intelligence; crops monitoring; machine learning; precision agriculture; remote sensing; smart agriculture; smart farming; unmanned aerial vehicles

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1. Introduction

With the increased demand for global food production and the effects of climate change, the modernization of agricultural practices has become inevitable. Agriculture that depends on traditional farming methods has remained reliable but is no longer sufficient to meet the demands of global food security. With increasing progress in unmanned aerial vehicles (UAVs), artificial intelligence (AI), and machine learning (ML), precision agriculture seems to be a promising approach for data-driven, efficient, and sustainable farming [1–4]. More importantly, AI and ML models such as convolutional neural networks (CNNs) and support vector machines (SVMs) further energize this type of analysis of the data they generate, making it possible to conduct critical agricultural tasks such as crop disease identification, yield forecasting, and resource allocation [5,6]. By simplifying complex analyses, AI integration provides predictions and insights into crop health and yields. Recent studies have suggested that AI-based unmanned aerial vehicle systems can detect crop diseases with a mean accuracy of over 90%, rendering such a tool necessary for modern agriculture [7]. An overview of the sensor, UAV, and AI technologies used in precision farming is shown in Figure 1.

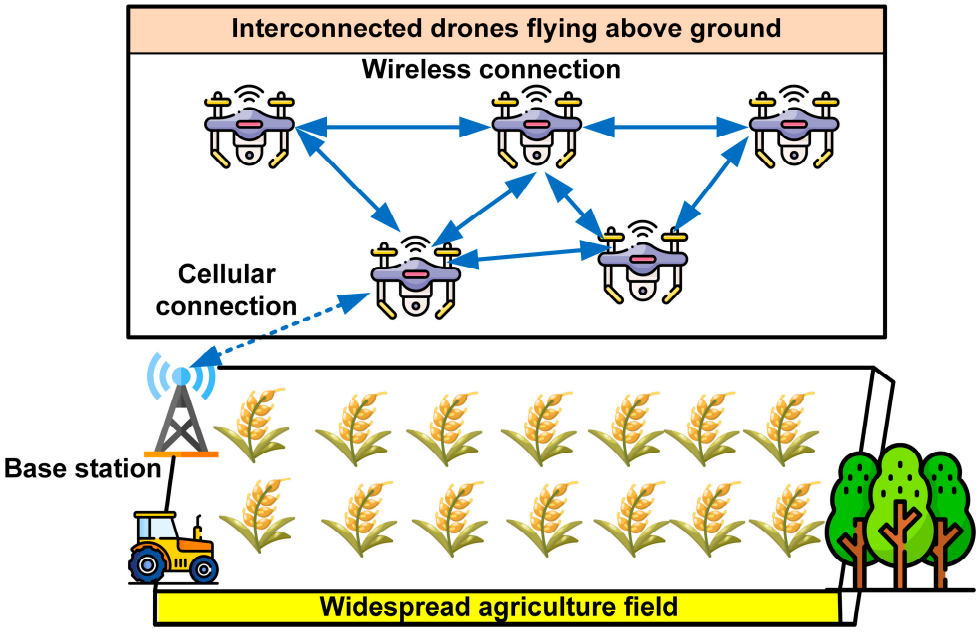


Figure 1. Using sensors, UAVs, and AI technology for precision farming.

Despite their transformative potential, the adoption of UAVs and AI in agriculture suffers from issues such as high initial costs, the need for technical expertise, and the environmental footprint of UAV operations [8,9]. UAVs have been integrated with AI for precision agriculture to maximize crop management, increase productivity, and reduce environmental impact. UAVs capture large amounts of high-resolution data, including multispectral, hyperspectral, and thermal images. AI algorithms are critical for processing these data. The application of deep learning and machine learning models helps analysts analyze crop health, identify pests, and assess soil conditions with a high degree of accuracy. UAV data that require high-resolution processing require substantial computational power and storage, particularly when dealing with multispectral and thermal data. Real-time processing may be required for AI models, which require either high-performance on-board computing capabilities or reliable connectivity to remote processing facilities. UAVs are restricted by battery limitations in terms of flight time, which limit their ability to collect data during a single flight. Investing in UAVs and AI technology can be expensive; however, long-term savings and productivity gains can be substantial. In precision agriculture, the use of UAVs and AI enables data-driven decision making and improves resource efficiency. With the development of this technology and increasing accessibility, agricultural productivity and sustainability are likely to increase. This paper reviews the current applications of UAVs and AI in precision agriculture, discussing their strengths, gaps, and proposals on how to solve these challenges. The paper proceeds to future research avenues—sensor technology, energy efficiency, and data integration—along with addressing ethical dilemmas regarding UAV usage in agriculture.

1.1. Contributions of This Study

The novel contribution of this study to the field of precision agriculture lies in its comprehensive survey of UAV-AI integration, focusing on the following areas:

Contribution and novelty: This study provides a comprehensive comparison between different types of UAVs and sensors, such as multispectral, hyperspectral, and thermal sensors, for crop monitoring, pest detection, and resource management. The novelty of this study ranges from the traditional models of UAVs to emerging ones, including hybrid or tethered drones, making the understanding of their applications in agriculture even broader.

AI and ML algorithms for precision farming: This study delves into AI and ML algorithms, including CNNs, SVMs, and decision trees, which process UAV-captured data for critical agricultural tasks such as disease detection, yield prediction, and irrigation management. The novelty lies in its comparative analysis of AI techniques, which points out their strengths and gaps and provides unique insights into which algorithms are most effective in performing specific agricultural tasks and how they improve precision farming.

AI-driven UAV innovations: A unique feature of this study is its detailed overview of existing AI-driven UAV innovations in precision agriculture, along with a discussion of their features, limitations, and performance metrics.

UAV adoption barriers and solutions: This paper highlights significant barriers to UAV adoption, such as UAVs being too expensive to use, having a short battery life, requiring technical expertise, and being associated with a regulatory environment that can be quite daunting. The novelty lies in presenting detailed practical solutions, ranging from sensor miniaturization for solar-powered UAVs to training programs, and specific regulatory recommendations that offer a clear roadmap for addressing these challenges in precision agriculture.

Challenges in existing solutions: This study evaluates the technical and operational challenges associated with the implementation of UAV and AI technologies in agriculture, recommending possible solutions for sensor miniaturization, improved energy efficiency, and autonomous flight systems.

Future research directions: This paper discusses the future of precision agriculture, where advanced AI techniques such as reinforcement learning and generative adversarial networks can be integrated into UAVs with better adaptability and operational efficiency.

1.2. Organization of This Paper

This review is organized into five sections, as shown in Figure 2. The remainder of the paper is outlined below.

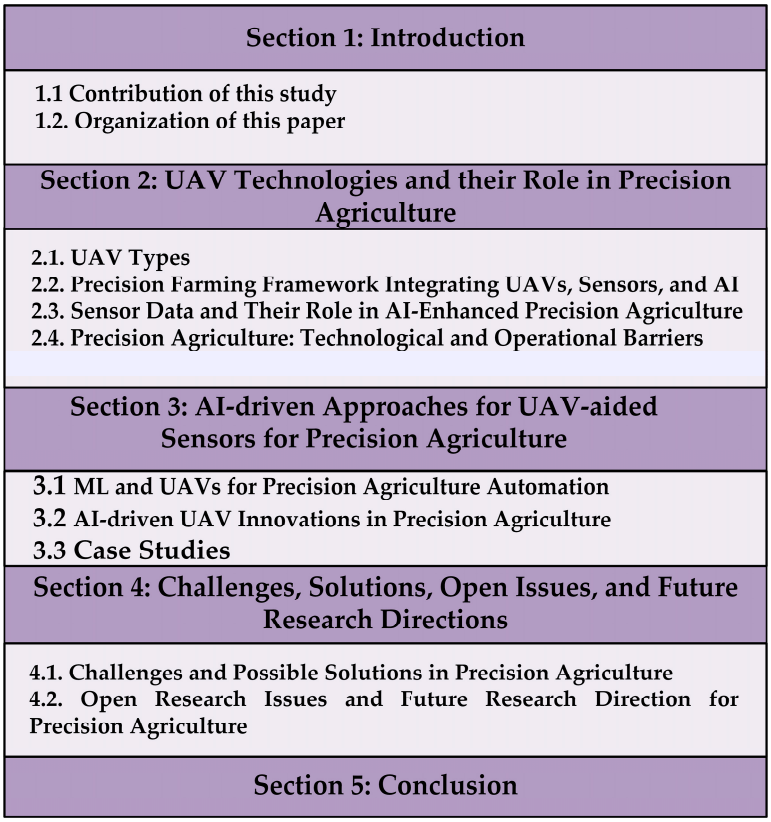


Figure 2. Outline of the paper.

In the next section, we present various UAV technologies and their role in precision agriculture. Afterward, we review recent AI-driven UAV-aided sensing techniques in precision agriculture. Following this, we present open research issues, challenges, and future research directions. Finally, we conclude the study in the next section.

2. UAV Technologies and Their Role in Precision Agriculture

UAVs are one type of aircraft capable of flying autonomously in the air without the assistance of a pilot, and their movement may be controlled remotely by an operator. UAVs are equipped with sensors and cameras that record and transmit images to operators [10].

2.1. UAV Types

UAVs have become significantly important for farming operations because they are widely used for crop monitoring, tracing infections, and irrigation management. UAVs can be further categorized into different classes depending on their design, payload, altitude range, sensor types, flight mode, and coverage area [11]. The general use of fixed-wing UAVs is to carry out large-area field mapping and crop health monitoring using a vector-P UAV [12]. This type of UAV has a longer endurance of up to 6 h and cruises at a higher altitude than the others, at up to 210 m [13]. In general, it carries sensors such as multispectral and thermal cameras, which help with crop variability and water resource management [14]. According to [15], rotary-wing UAVs are more agile because they can perform tasks that require precise maneuverability, such as close-range monitoring and disease identification in specific crops such as citrus. The endurance time of these sensors is generally 10–20 min; they are generally more versatile for use in small- to medium-sized spaces with multispectral and thermal sensors [16]. Specialized UAVs include vertical takeoff and landing (VTOL) drones, which offer even more advanced applications in variable-rate fertilization and automated crop health assessment [17].

Several types of UAVs that are used in agriculture are shown in Figure 3. There are three basic types of UAVs based on their aerodynamic characteristics: fixed-wing, rotary-wing, and hybrid UAVs. Compared with rotary-winged and hybrid UAVs, fixed-wing UAVs have better aerodynamic efficiency, resulting in a longer flight duration. In the context of agriculture, this provides advantages for mapping large fields, surveying large areas, and monitoring crop health. Owing to their higher speeds and longer ranges, they are ideal for farms requiring monitoring over a large area, including tens or hundreds of acres. A fixed-wing UAV is generally more cost-effective over long distances because it requires less power to sustain flight, which makes it an ideal tool for extensive land surveys and crop analysis. Fixed-wing UAVs cannot hover in place, which limits their application to tasks that require stationary positioning, such as inspecting specific plants or applying spot treatments.

Rotary-wing UAVs are ideal for tasks that require precision, such as monitoring specific plants, administering localized treatments, or conducting detailed inspections. Owing to its small takeoff and landing space requirements, this type of UAV is well suited to fields with uneven or confined terrains. A rotary-wing UAV consumes less energy, resulting in shorter flight times and a reduced range, limiting its efficacy for surveys in large areas. Typically, these UAVs travel at a slower pace, making them less effective for quickly covering large areas. A hybrid UAV combines the benefits of both fixed-wing and rotary-wing UAVs, often integrating VTOL capabilities with aerodynamic efficiency. Due to their vertical takeoff and landing capabilities, hybrid UAVs can operate in confined areas. They can also achieve longer flight times than fixed-wing UAVs when compared with rotary-wing UAVs. With the combination of hovering and efficient forward flight, they are capable of handling both large-area surveys and precision tasks, as required. Compared with simpler UAV types, hybrid UAVs are generally more complex to design and operate, increasing their cost and making them prone to mechanical failure.

Several types of sensors are installed on the UAV to sense data from the crops. Hyperspectral, light detection and ranging (LIDAR), and thermal imaging systems on sophisti-

cated sensors provide real-time data to optimize agricultural inputs, boost crop productivity, and reduce adverse environmental effects [18–20]. For example, the Yamaha aero robot R-50 (Yamaha Motor Company, Iwata, Japan), Yamaha KG-135 (Yamaha Motor Company, Iwata, Japan), and YH300 (Yamaha Corporation, Hamamatsu, Japan) have a payload ability for heavy-duty applications such as pesticide spraying up to 22.7 kg and a convincing endurance of 5 h, making them suitable for large-area coverage [21,22]. Table 1 summarizes the types of UAVs and their applications in precision agriculture.

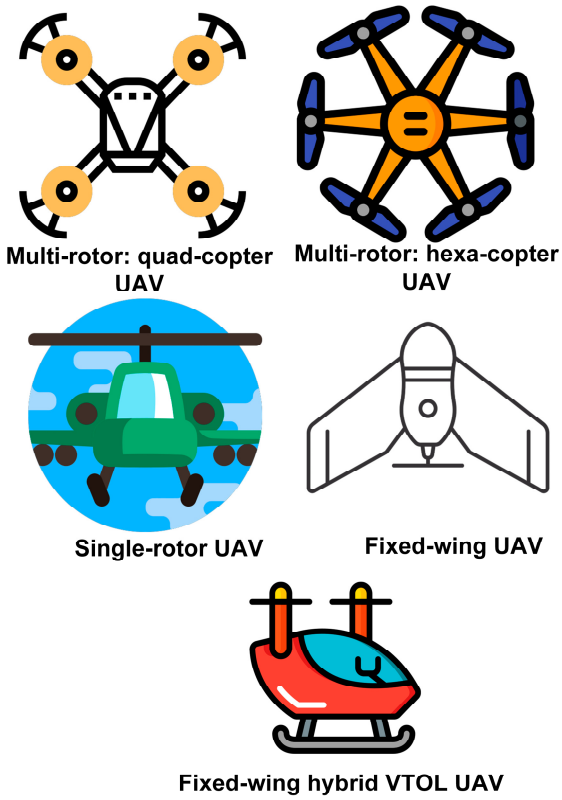


Figure 3. A variety of UAVs are used in agriculture.

Table 1. Types of UAVs and their applications in precision agriculture.

Type of UAV	Application in Precision Agriculture	Payload	Altitude	Endurance	Type of Sensor Used	Flight Mode	Coverage Area
Fixed-wing UAV [23]	Crop monitoring, large-field mapping, and irrigation	2–5 kg	100–500 m	1–4 h	Multispectral, RGB, and thermal	Autonomous	Large areas
Multi-rotor UAV [23]	Crop scouting, pest monitoring, and variable-rate application	1–3 kg	50–200 m	20–40 min	Multispectral, RGB, and LIDAR	Manual/ autonomous	Small/ medium areas
Hybrid UAV [23]	Field mapping, crop health monitoring, and precision spraying	2–10 kg	100–400 m	2–5 h	Multispectral, RGB, and thermal	Autonomous	Large areas

Table 1. *Cont.*

Type of UAV	Application in Precision Agriculture	Payload	Altitude	Endurance	Type of Sensor Used	Flight Mode	Coverage Area
VTOL UAV [24]	Crop inspection on uneven terrain and variable-rate fertilization	3–8 kg	50–400 m	1–2 h	Multispectral, RGB, and hyperspectral	Autonomous	Medium/large areas
Nano/mini UAV [25]	Close-range monitoring, disease detection, and small-farm inspection	0.1–0.5 kg	10–100 m	10–20 min	RGB and multispectral	Manual/autonomous	Small areas
Autonomous UAV [26]	Precision irrigation, spraying, and automated crop health analysis	5–15 kg	50–300 m	3–6 h	Thermal, hyperspectral, and LIDAR	Fully autonomous	Large areas
Tethered UAV [27]	Continuous monitoring and real-time data collection	Up to 10 kg	20–100 m	Unlimited (Tethered)	Multispectral and RGB	Tethered	Fixed position
Fixed-wing UAV [28]	Crop monitoring and identification of crop response	Small camera	90 m	<1 h	Thermal infrared and thermal camera	Autonomous	Large areas
Field copter UAV [29]	High-resolution image acquisition and water status in vineyards	<1 kg	Low altitude	Not available	Multispectral and thermal cameras	Manual/autonomous	Small areas
Multi-rotor UAV [30]	Crop supervision and citrus disease identification	<1 kg	100 m	10–20 min	Six-channel multispectral and multi-band	Manual/autonomous	Small areas
Hexa-copter UAV [31]	Cultivation investigation and multispectral data processing	Up to 2.5 kg	Low altitude	12–20 min	Tetracam camera	Autonomous	Medium areas
Fixed-wing UAV [32]	Grain crop attribute assessment	<1 kg	Low altitude	<1 h	Digital camera with image sensor	Autonomous	Medium areas
Vario XLC helicopter [33]	Weed management and reduction of herbicide use	7 kg	Low altitude	30 min	3D multispectral imaging sensors	Manual/autonomous	Small areas

2.2. Precision Farming Framework Integrating UAVs, Sensors, and AI

Precision farming primarily uses UAVs for agricultural productivity optimization. It is fitted with advanced imaging sensors and cameras, producing high-resolution images that can assist farmers with crop health monitoring, determine the health status of the soil, and even detect pests or nutrient deficiencies in soils [34,35]. Real-time data allow for precise decision making with the least consumption of excess fertilizers, pesticides, and water, saving money and protecting the environment [36]. Crop monitoring is a major application of UAVs; there are multispectral and hyperspectral sensors used in crop vigor and chlorophyll content appraisal, together with assessing stress in the plant [37,38].

UAVs utilize such sensors to provide highly detailed maps of variations in plant health in the field. This can then be used to facilitate actions in specific places that improve the quality and efficiency of the yield. Figure 4 depicts various applications of UAVs in precision agriculture.

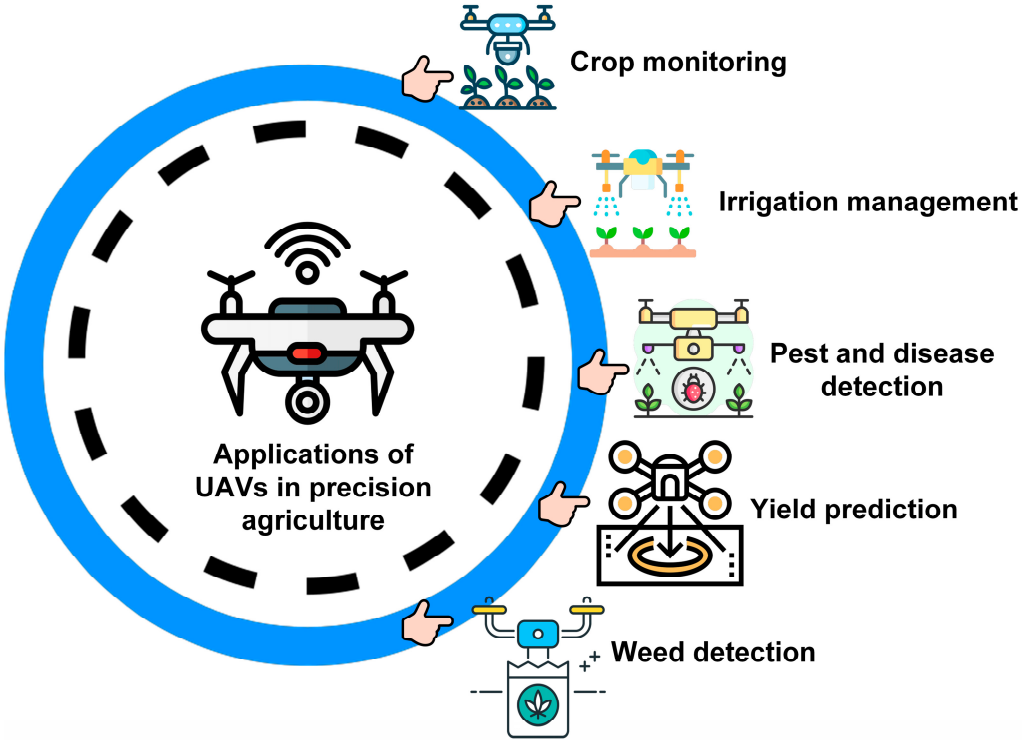


Figure 4. Applications of UAVs in precision agriculture.

Fixed-wing UAVs are suitable for vast spans of farmland, as they cover longer ranges and exhibit endurance over long distances [39]. Rotary-wing or multi-rotor UAVs have a greater degree of maneuverability and could be better used where accuracy needs to be met in applications such as plant inspection or localized treatment [40]. UAVs allow for mapping and surveying, along with the creation of high-resolution 3D models of farmland for topographic studies, which are essential for planning irrigation and controlling erosion [41]. With the further development of AI and automation, UAVs are likely to enhance autonomous decision making in precision farming, thereby augmenting crop productivity and sustainability [42].

The diagram in Figure 5 illustrates the integration of UAVs, sensors, and AI in precision farming, highlighting the stages involved in collecting, processing, analyzing, and utilizing agricultural data for informed decision making.

Data collection: Sensors (for soil, weather, and crop health), satellites, and drones are used for gathering data. Different types of cameras, such as multispectral, visible, and thermal cameras, are employed to capture detailed environmental and crop data.

Data processing: The calibration and noise removal of all raw data are performed. Information fusion techniques merge information from different sources into complete datasets.

Feature extraction: The processed data undergo feature extraction, which includes the vegetation indices of normalized difference vegetation index (NDVI) and normalized difference red edge index (NDRE). Texture analysis, object detection, and other important attributes help in crop monitoring.

ML algorithm: Machine learning algorithms are applied to the extracted features using approaches such as supervised learning (classification and regression), unsupervised learning (clustering) [43], and reinforcement learning [44].

Model training: The data are split into training, validation, and testing sets. Techniques such as hyperparameter tuning and cross-validation are applied to optimize the performance of the models.

Model evaluation: The models are evaluated based on metrics such as accuracy, precision, recall, and F1 score to determine their effectiveness.

AI-driven UAV operations: The AI models are deployed in UAV systems to automate real-time data collection, analysis, and decision making from aerial perspectives. These UAVs dynamically adapt to changing field conditions to optimize precision farming processes.

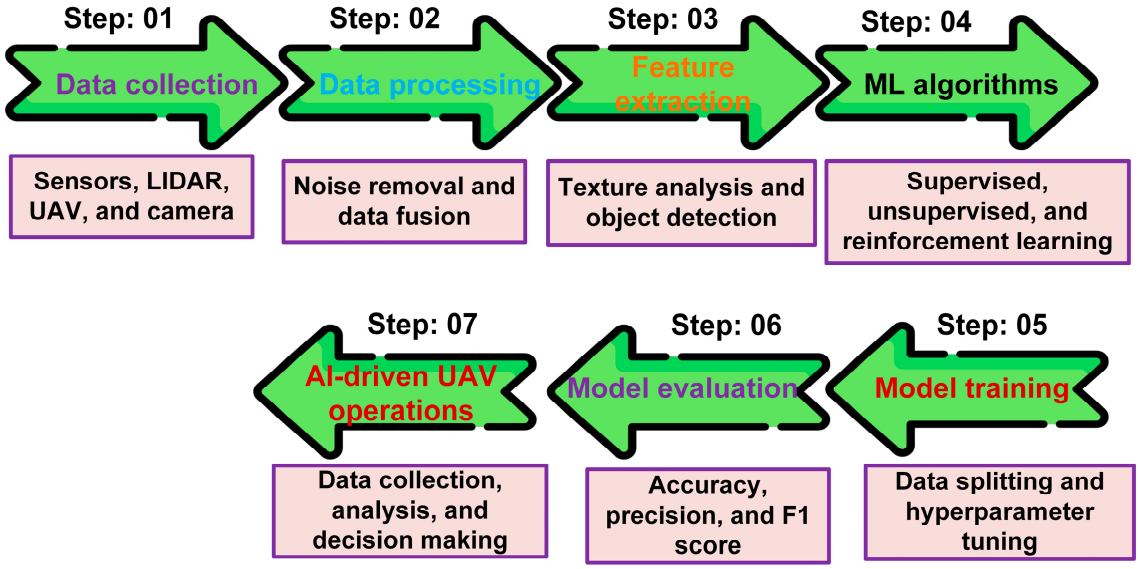


Figure 5. Workflow for integration of UAVs, sensors, and AI in precision farming.

2.3. Sensor Data and Their Role in AI-Enhanced Precision Agriculture

The quality of sensor data plays a significant role in precision agriculture to increase the accuracy and efficiency of the decisions made by AI. Critical sensor performance metrics, such as resolution, spectral range, and amount of data, influence the extent to which AI models analyze, predict, and optimize agricultural operations [45]. Such high-quality sensor data allow for the accurate monitoring of crops, identification of diseases, and management of resources, ultimately resulting in better yield prediction and sustainable farming practices. Therefore, understanding the relationship between the performance of sensors and AI outcomes is important for unlocking the full potential of precision agriculture technology [46]. Table 2 illustrates the effect of sensor performance metrics and their influence on AI in precision agriculture applications.

Table 2. Sensor performance metrics and their influence on precision agriculture.

Sensor Type	Key Metrics	Agricultural Application	AI-Driven Model Performance	Limitations
RGB camera [47]	Resolution and frame rate	Crop growth monitoring and disease detection	Higher resolution provides detailed imagery and improves image classification tasks (e.g., identifying disease spread patterns)	Limited spectral range cannot capture chlorophyll content or subtle crop health changes
Multispectral sensor [48]	Spectral range (visible to near-infrared) and NDVI	Vegetation health monitoring and weed detection	Enables the calculation of vegetation indices like NDVI and enhances AI models' predictions of plant health and vigor	Sensitive to atmospheric conditions (e.g., cloud cover and light intensity)

Table 2. *Cont.*

Sensor Type	Key Metrics	Agricultural Application	AI-Driven Model Performance	Limitations
Hyperspectral sensor [49]	Spectral resolution and data volume	Nutrient monitoring and disease identification	Higher spectral resolution improves accuracy in identifying nutrient deficiencies and crop diseases and is critical for detailed AI/ML classification	Generates large datasets that can slow down AI/ML model processing and is costly to deploy
Thermal camera [50]	Temperature sensitivity and spatial resolution	Irrigation management and water stress detection	Provides precise thermal data and allows AI models to optimize irrigation by identifying water stress in crops earlier	Lower spatial resolution and less effective for small field details compared to RGB or multispectral
LIDAR [51]	Point density and range	Topographic mapping and biomass estimation	Produces precise 3D models of terrain and biomass and improves AI-based yield prediction and terrain mapping accuracy	Limited use in densely vegetated areas and expensive equipment
Synthetic aperture radar (SAR) [52]	Penetration depth and resolution	Soil moisture monitoring and all-weather mapping	Useful for all weather and all light conditions and supports AI models in texture and moisture analysis with improved performance in diverse weather situations	Requires high technical expertise for accurate interpretation and high operational costs

Precision agriculture relies heavily on sensors, as they provide data that assist farmers in making informed decisions regarding crop management, irrigation, pest control, and other aspects of agriculture. Table 3 provides an overview of several real-world conditions that may limit the effectiveness of these sensors.

Table 3. Limitations of sensors and the potential of sensor fusion in precision agriculture.

Aspect	RGB Sensors	Multispectral Sensors	Thermal Sensors	LIDAR Sensors
Weather conditions	Affected by cloudy weather and light variations, reducing image quality [19].	Sensitive to cloud cover, causing inconsistent data collection [40].	Less impacted by clouds, but extreme heat can distort readings [31].	Unaffected by weather, providing reliable 3D models [42].
Light conditions	Perform poorly in low light, especially during early morning or late afternoon [26].	Dependent on sunlight; struggles in low-light and shadowed areas [41].	Effective in low light, using temperature differences to detect crop stress [19].	Independent of lighting; capture accurate terrain models anytime [31].
Sensor calibration	Require frequent recalibration, especially with varying light conditions [42].	Calibration needed to maintain accuracy of vegetation indices [40].	Minimal calibration required; errors may distort temperature readings [41].	Require minimal calibration, terrain data generally robust [31].
Operational challenges	Susceptible to wind and UAV vibrations, reducing image accuracy [42].	Vibrations distort readings, affecting vegetation indices [19].	Less affected by UAV movement; reliable for detecting water stress [40].	Minimal impact from wind or vibrations, highly reliable for 3D mapping [41].
Single-sensor limitations	Limited to surface imagery, cannot assess below the canopy [41].	Provide vegetation indices but lack soil moisture data and structural insights [40].	Detect water stress but lack detailed surface analysis [19].	Excellent for terrain data but lack vegetation health detection [42].
Sensor fusion potential	Thermal and RGB improves accuracy by balancing light-dependent limitations [19].	Multispectral and LIDAR compensate for weather and lighting variations, providing comprehensive crop and terrain data [41].	Combining thermal with multispectral data enhances irrigation insights and water stress detection [40].	LIDAR with multispectral/thermal offers robust yield estimation and biomass mapping [31].

2.4. Precision Agriculture: Technological and Operational Barriers

Although UAVs bring many benefits to precision agriculture, there are still several technological and operational barriers that hinder their widespread adoption. This may be due to high investment costs, technical expertise, and regulatory constraints [53]. All of these must be mapped to allow UAV technologies to be fully unleashed in modern farming practices. Table 4 illustrates the technical and operational barriers to the adoption of UAVs in precision agriculture.

High initial cost: The high front-end costs of UAVs, sensors, and processing software are major inhibitors, especially for small- and medium-sized farms. High-cost advanced sensors such as multispectral and LIDAR, as well as cloud-based processing systems that are expensive to lease, limit wider adoption, and even government subsidies or leasing programs cannot fully mitigate cost as a limitation, primarily for smaller farms [54].

Limited battery life: UAVs typically fly for only 20–40 min, and heavier payloads reduce the flight time even more. This means that frequent battery replacement in large fields is always required [55]. Solutions such as solar-powered UAVs are in development, but they are not yet widely available, thereby reducing operational efficiency.

Regulatory constraints: Strict regulations limit UAV flight altitudes, require line-of-sight operations, and raise privacy concerns. Such regulations minimize UAV flexibility and may delay adoption in regions with strict airspace rules. Lobbying for agricultural exemptions and standardized safety guidelines may ease some restrictions.

Technical expertise: Operating UAVs and analyzing the collected data require considerable expertise in geographic information systems (GIS) and machine learning, which acts as a barrier to uptake, especially among farmers without technical backgrounds. Although AI-driven, user-friendly, and more packaged software solutions are being developed, the reliance on consultants is still massive, increasing costs.

Environmental impact: The battery discharge and noise of UAVs may prove harmful to the environment, especially if they are to be implemented in ecologically sensitive areas. Although sustainable technologies such as hydrogen fuel cells can minimize the footprint in the environment, adaptations in sensitive regions can be affected by the present limitations in battery power.

Table 4. A summary of technical and operational barriers to the adoption of UAVs in agriculture.

Factor	Description	Technological Constraints	Sensor Integration	Data Processing Challenges	Potential Solutions	Impact on Adoption
High initial cost [56]	For small and medium-sized farms, purchasing UAVs, sensors, and supporting software can be prohibitive.	Expensive hardware (UAVs, sensors, and processing systems).	Multispectral, thermal, and LIDAR sensors significantly increase costs.	Processing large datasets requires expensive computing infrastructure or cloud-based systems, which increase the total cost.	Government subsidies, leasing programs, or collaborative buying of models among farmers to reduce the financial burden.	Limits widespread adoption, especially for small farms with limited capital.
Limited battery life [57]	Most UAVs have short flight times, often requiring frequent battery changes, which limits their use in larger agricultural fields.	Short battery life (20–40 min), frequent need for recharging or replacement.	Heavier payloads like LIDAR or thermal cameras reduce flight time further, exacerbating the battery issue.	Real-time data transmission is difficult in larger fields; UAVs may need to land frequently, interrupting continuous data collection.	Development of energy-efficient UAVs, use of solar-powered drones or hybrid UAVs with longer endurance.	Decreases operational efficiency in large fields, increasing downtime and labor costs.

Table 4. *Cont.*

Factor	Description	Technological Constraints	Sensor Integration	Data Processing Challenges	Potential Solutions	Impact on Adoption
Technical expertise [58]	Operating UAVs and interpreting the collected data requires significant technical expertise, which may not be readily available to all farmers.	Need for specialized training in UAV operation and data analysis software (GIS, machine learning tools).	The integration of complex sensor suites (multispectral, thermal) requires advanced calibration and data preprocessing skills.	Data analysis often requires machine learning algorithms that need tuning and training, which is complex for users without technical backgrounds.	User-friendly software interfaces, AI-driven automated data processing, and training programs for farmers.	Limits adoption by non-expert users; dependency on specialized service providers or consultants increases operational costs.
Regulatory constraints [59]	UAV use is regulated in many regions, limiting flight altitude, operating areas, and privacy concerns.	Flight restrictions in many countries (maximum altitude limits and requirement for line-of-sight operation).	N/A	Compliance with aviation regulations requires software and manual input to ensure safe operating boundaries, which can reduce flight flexibility.	Lobbying for agricultural exemptions in UAV regulations, creating operational safety standards for agricultural drones.	Regulatory uncertainty can delay or prevent widespread adoption, especially in regions with strict UAV regulations.
Environmental impact [60]	UAV operations may have an environmental footprint, especially with battery disposal, wildlife disturbance, and noise pollution.	Limited battery life leads to frequent disposal or recycling issues with lithium-ion batteries.	N/A	Processing in remote locations may require additional infrastructure (generators and signal towers), impacting ecosystems.	Development of sustainable battery technologies (hydrogen fuel cells) and more eco-friendly UAV designs.	Environmental concerns may slow adoption in areas with sensitive ecosystems or strict environmental regulations.

3. AI-Driven Approaches to UAV-Aided Sensors for Precision Agriculture

With agricultural technology, sensor data analysis developed with UAVs has become highly integrated across precision agricultural systems. For example, such development offers real-time monitoring for the critical management of agricultural parameters, such as crop health, soil conditions, and water stress levels. Sensors include RGB, multispectral, hyperspectral, and thermal imaging sensors. These methods collect enormous amounts of data, which deep learning algorithms analyze to classify, regress, and detect objects. A good example of this would be a multispectral and hyperspectral soil and crop condition assessment, which is possible because it allows AI models to predict water content, salinity, and health conditions in crops [61]. Crop water stress can be monitored further using thermal sensors that strengthen irrigation scheduling decisions [62].

AI-based yield prediction ensures that farmers can harvest when they have maximum produce and make other informed decisions [63]. These data sources are revolutionizing precision agriculture by minimizing manual labor input and improving decision-making accuracy, thereby optimizing agricultural practices. In this regard, incorporating input from AI-driven techniques into handling UAV and sensor data would rejuvenate farming and data-related activities toward becoming more sustainable. Table 5 shows the use of different AI-driven techniques to process UAV-aided sensor data for several specific agricultural applications.

Table 5. AI techniques for analyzing UAV-aided sensor data in agriculture.

Elements	Representative Examples	UAV Sensors	AI Problems	Altitude	Endurance	Flight Mode
Area estimation [42]	Soybeans and several vegetation types	RGB, multispectral, and hyperspectral	Classification	100–500 m	1–3 h	Autonomous
Soil conditions [15]	Water content and salinity	Multispectral, hyperspectral, and thermal	Regression	50–200 m	20–40 min	Manual/ autonomous
Crop parameters [64]	Crop height, canopy cover, biomass, leaf area index (LAI), and leaf nitrogen content	RGB, multispectral, hyperspectral, and LIDAR	Regression and classification	50–400 m	1–4 h	Autonomous
Crop diseases [65]	Yellow rust, citrus canker, bacterial wilt, foliage disease, and late blight	RGB, multispectral, and hyperspectral	Regression and classification	50–300 m	2–5 h	Manual/ autonomous
Crop pests [66]	Fruit pests, sheath blight, and soybean pests	RGB, multispectral, and hyperspectral	Object detection and classification	20–100 m	20–60 min	Fully autonomous
Crop water Stress [67]	Barley, maize, vineyard, and wheat	RGB, multispectral, and thermal	Index-based analysis, and regression	50–500 m	1–3 h	Autonomous
Weeds [68]	Blackgrass in wheat, weeds in sugar beet and soybean	RGB, multispectral, and hyperspectral	Classification and object detection	50–200 m	1–2 h	Autonomous
Yield prediction [69]	Soybean, cotton, citrus, and oil palm fruit	RGB, multispectral, and stereo camera	Regression and object detection	100–400 m	1–3 h	Fully autonomous
Harvest decision [70]	Soybean, peanut, broccoli, and canola	RGB, multispectral, and hyperspectral	Regression and object detection	50–300 m	2–5 h	Manual/ autonomous
Irrigation scheduling [71]	Sugarcane, wheat, and rice	RGB and thermal	Index-based analysis and regression	50–150 m	20–60 min	Autonomous
Nutrient monitoring [72]	Maize, sorghum, and wheat	Hyperspectral and multispectral	Classification and regression	100–500 m	1–3 h	Autonomous
Crop growth stage [73]	Cotton, maize, and wheat	RGB and multispectral	Classification and object detection	50–200 m	1–2 h	Fully autonomous
Post-harvest monitoring [74]	Cereals, fruits, and vegetables	RGB, multispectral, and hyperspectral	Regression and classification	50–300 m	1–4 h	Autonomous
Early disease detection [75]	Rice, wheat, and tomato	Thermal, hyperspectral, and RGB	Object detection and classification	50–400 m	1–3 h	Manual/ autonomous

3.1. ML and UAVs for Precision Agriculture Automation

The application of ML and UAVs in precision agriculture has revolutionized traditional farming methods, enabling more efficient and data-driven agricultural practices. Using advanced sensors combined with very sharp ML algorithms, UAVs can perform several critical operations such as plant recognition, disease identification, weed identification, crop yield prediction, and soil nutrient mapping. These innovations lead to higher precision, quicker processing periods, and better resource utilization, among many others, in modern agriculture. However, other challenges, such as environmental noise, computational demands, and UAV battery limitations, still pose obstacles that require refinements in the design of such technologies. An overview of agricultural automation is presented in Table 6.

Table 6. Summary of agricultural automation using UAV, sensors, and ML for various applications.

Agricultural Tasks	Crop/System	Contributions	Automation Vehicle/ Platform	ML Algorithms	Performance Metrics	Strengths	Gaps
Plant classification	Grapes and orchards [18]	A UAV is used for fast and accurate plant classification	UAV with RGB sensors	CNN and random forest	Accuracy: 92% and recall: 90%	High accuracy with RGB sensors	High computational demands, real-time processing challenges
	Wheat and barley [76]	Enhanced plant growth analysis with UAVs and ML	UAV with multispectral sensors	Deep learning and SVM	Accuracy: 94% and precision: 92%	Higher accuracy with multispectral sensors	Sensitive to lighting variability
	Soybean and corn [15]	LIDAR systems improve classification accuracy	UAV with LIDAR and near-infrared (NIR) cameras	Decision trees and SVM	Accuracy: 91% and processing time: 2 sec/field	Fast processing time (2 sec/field)	Calibration complexities may limit accuracy
Disease detection	Olive and citrus [77]	Accurate disease detection in early stages	UAV with thermal and hyperspectral cameras	YOLO and CNN	Detection accuracy: 90% and specificity: 88%	High accuracy in disease monitoring	Environmental noise and temperature changes may affect accuracy
	Multiple crops [78]	Transfer learning improves plant disease detection accuracy	UAV with RGB and hyperspectral cameras	Transfer learning and CNN	Detection accuracy: 95% and precision: 92%	High accuracy and minimal training data required	Sensitive to variation in input data
	Tomato and cucumber [79]	Early stage disease detection with UAVs and ML	UAV with RGB and multispectral sensors	SVM	Accuracy: 88% and F1-score: 85%	Reliable F1-score (85%)	Environmental factors (e.g., weather) may impact detection
	Fruit trees [80]	Thermal cameras help detect invisible disease symptoms	UAV with RGB and thermal cameras	Neural networks	Detection accuracy: 93% and precision: 90%	High precision (90%)	UAV flight time and battery life limitations

Table 6. *Cont.*

Agricultural Tasks	Crop/System	Contributions	Automation Vehicle/ Platform	ML Algorithms	Performance Metrics	Strengths	Gaps
Weed detection	Row crops and corn [18]	Precise weed detection with hyperspectral imaging	UAV with hyperspectral imaging	k-nearest neighbor and decision trees	Weed identification accuracy: 93% and recall: 90%	Accurate weed identification	Environmental and lighting interference issues
	Rice and soybean [79]	High-accuracy weed detection with ML	UAV with multispectral cameras	Random forest and SVM	Detection accuracy: 91% and F1-score: 88%	High detection accuracy (91%)	Requires consistent lighting for optimal performance
Crop yield prediction	Corn and rice [79]	Supports real-time yield estimation	UAV with multispectral sensors	Random forest and XGBoost	Yield prediction accuracy: 85% and error: 1.02	Effective in yield prediction	Yield prediction accuracy can vary due to inconsistent data availability
	Wheat and grapes [81]	Improves harvest decisions with UAV data	UAV with RGB and multispectral sensors	Support vector regression	Prediction accuracy: 86% and error: 0.9	Effective RMSE score for prediction	Yield prediction accuracy varies by crop type
	Grapes [82]	Vineyard LAI prediction using CNN	UAV with RGB and LIDAR sensors	ResNet50 and Xception	Error: 0.28 for LAI estimation	Accurate LAI prediction with Xception model	Models had inconsistent performance in some fields
Soil nutrient mapping	Soybean and corn [80]	Detailed soil nutrient mapping with UAV-LIDAR	UAV with LIDAR and NIR sensors	Regression algorithms	Soil composition accuracy: 86% and precision: 84.5%	High precision for soil composition analysis	Limited resolution may hinder fine-scale mapping
	Potato and wheat [18]	Enhanced mapping of soil nutrients	UAV with RGB and multispectral cameras	Decision trees and random forest	Soil nutrient mapping accuracy: 87% and recall: 85%	High accuracy (87%) in soil nutrient mapping	UAV flight time and coverage area are limiting factors
Pest management	Citrus and grapevine [83]	Supports automated pest detection	UAV with high-resolution cameras	Neural networks and YOLO	Pest detection accuracy: 91% and precision: 88%	High accuracy (91%) in pest identification	Limited flight time and rough terrain may interfere with UAV performance
	Apple and pear [84]	Detection of apple lesions using YOLO models	UAV with RGB and thermal cameras	Cycle GAN and dense net	F1-score: 81.6%	Real-time detection of apple lesions	Limited accuracy for occluded objects
	Corn and cotton [79]	Effective in identifying and managing pests	UAV with multispectral cameras	Deep learning and random forest	Pest identification accuracy: 89% and precision: 87%	High precision in pest identification	Environmental noise may affect accuracy during operation

Table 6. *Cont.*

Agricultural Tasks	Crop/System	Contributions	Automation Vehicle/ Platform	ML Algorithms	Performance Metrics	Strengths	Gaps
Autonomous spraying	Vineyard and wheat [15]	High-precision spraying reduces chemical use	UAV with spraying mechanisms	Reinforce-ment learning	Spray precision: 94% and coverage efficiency: 87%	High precision and efficiency in spraying operations	Operational complexity can be a barrier to widespread adoption
	Cotton and corn [81]	Improves efficiency in pesticide application	UAV with autonomous spraying systems	Deep learning and XGBoost	Spray precision: 91% and coverage efficiency: 90%	High spray precision and coverage	Terrain limitations and battery life reduce operational time

3.2. AI-Driven UAV Innovations in Precision Agriculture

Precision agriculture has been revolutionized through the infusion of AI and UAVs, which enables farmers to perform more precise tasks with information-driven decisions and high efficiency. UAVs are supplied with AI and ML algorithms that include crop monitoring in real time, crop health analysis, weed and pest detection, and optimal irrigation practices. Current advancements allow UAVs to have integrated multispectral, thermal, and LIDAR sensors alongside crop health conditions within the soil, hence enabling intense analysis for intervention and thus optimizing yield management. With AI-based platforms, it is possible to automate complex functions, such as precision seeding and the application of chemicals, including plant stress detection, as tools for minimizing labor input and maximizing resource use. Such systems scan through large areas of aerial imagery and sensor data while running in real time, using sophisticated machine learning algorithms to render on-the-spot decisions with high accuracy. However, challenges such as high initial setup costs, environmental interference, and limited flight times remain, requiring ongoing innovation to enhance scalability and usability globally for farmers. AI-driven UAV industrial solutions used for precision agriculture are listed in Table 7.

Table 7. AI-enabled UAV-based industrial solutions for precision agriculture.

Category	Product	Description	ML Algorithm	Features	Limitations	Performance Metrics
AI-enabled UAV for precision mapping	DJI Phantom 4 RTK [85]	UAV with real-time kinematic (RTK) technology for precision mapping and crop monitoring	CNN for image analysis	High-accuracy global positioning system, real-time crop monitoring, multispectral imaging, and RGB sensors	Expensive, battery life limitations in larger fields	RTK positioning accuracy: ±1 cm, flight time: 30 min, image resolution: 20 MP, and image accuracy: 98%
AI crop monitoring platform	Taranis [86]	AI-driven platform for crop health and pest monitoring using high-resolution aerial imagery	Deep learning and SVM for crop health analysis	Automated pest detection, early disease identification, and UAV-integrated multispectral sensors	Susceptible to weather conditions, reliant on drone availability	Detection accuracy: 95%, image resolution: 0.5 cm/pixel, and data processing speed: 30 min per hectare

Table 7. *Cont.*

Category	Product	Description	ML Algorithm	Features	Limitations	Performance Metrics
AI-driven seeding and spraying drone	XAG P100 agricultural drone [87]	AI-powered UAV designed for precision seeding and spraying in agriculture	Reinforcement learning for flight optimization and neural networks for seeding patterns	Autonomous flight, precision seeding, AI-driven spraying patterns, RGB and spraying mechanisms	Limited payload capacity, requires expert handling	Seeding accuracy: 95%, flight duration: 25 min, payload capacity: 25 kg, flight accuracy: 96%, and spraying accuracy: 94%
AI-enhanced smart irrigation	Net Beat by Netafim [88]	AI-based irrigation system that optimizes water usage based on real-time data	Regression models and decision trees for water optimization	Automated irrigation control, water usage optimization, moisture and flow sensors	High initial setup cost, requires advanced sensors	Water usage reduction: up to 30%, irrigation precision: ±3%, response time: real-time
Soil and plant stress monitoring UAV	CropX [89]	AI-based platform for soil and crop monitoring, integrating sensors for plant health detection	Random forest and k-nearest neighbors (KNN) for stress detection	Soil moisture analysis, automated stress detection, cloud-based insights, and soil moisture sensors	Data accuracy depends on sensor placement and limited coverage area	Plant health detection accuracy: 92%, moisture sensor precision: $\pm 0.03\ m^3/m^3$, and cloud processing speed: <5 min
Soil health monitoring system	Arable mark 2 [90]	Solar-powered sensor that tracks soil moisture, temperature, and plant health metrics	SVM for soil health prediction	Solar-powered, real-time soil health tracking, wireless connectivity, temperature and soil moisture sensors	Limited coverage area and expensive for large-scale fields	Sensor accuracy (moisture): $\pm 0.02\ m^3/m^3$, temperature accuracy: ±0.5°C, and soil moisture accuracy: 98%

3.3. Case Studies

In this section, we present a few real-life examples, where UAVs and AI have been used in precision agriculture.

UAVs in precision agriculture [21]: An Iowa corn farmer made some leaps in precision agriculture with the help of a UAV integrated with multispectral cameras. This development enabled them to observe their crops over an extensive area in detail regarding their crop health status. Applying pesticides through UAVs generated an extra income of approximately $434–488 per hectare and required much less time to apply pesticides.

Tethered UAVs for spraying and monitoring [22]: The utility of tethered UAVs lies in their practical applications, which can include crop monitoring and precision spraying, and they operate continuously because power is taken directly from the ground. This has been proven to enhance spraying accuracy and extend operational duration because tethered UAVs are not limited by the battery life of traditional UAVs. For example, on large farms, tethered UAVs can continually monitor pest control and nutrient application without interruption, thereby enhancing coverage. This method has applications in crop health monitoring and large-scale mapping. In applications where data need to be collected regularly, such as growth monitoring and health checks, tethered UAVs have been successfully applied because they can operate all day, unlike battery-powered drones. Case studies showed that farms using tethered UAVs for such applications reported higher data resolution because the UAVs could stay airborne for longer periods and capture high-quality imagery for detailed crop analysis. The effectiveness of the project is reflected in the reduction in fuel and labor costs, and tethered UAVs minimize the occurrence of

frequent manual interventions as well as refueling or recharging stops, particularly in large-field applications. Although savings in specific costs were not established in this study, the uninterrupted operation capability of tethered UAVs yields considerable labor-saving opportunities in terms of decreased downtime based on Asian and European commercial farm trials.

Hybrid UAVs in agriculture [23]: By leveraging precision agriculture techniques using UAVs, this case study discusses the pragmatic application of hybrid UAVs for multi-purpose work in the field. Hybrid UAVs boast the unique traits of being combined with those of fixed winged and multi-rotor UAV types; thus, they prove to suit diversified uses under agricultural management. When deployed in an intensive agricultural setting that covers diversified crops such as maize and wheat, hybrid UAVs would still retain the multispectral camera configurations for crop health mapping, as well as performing targeted applications of fertilizers and pesticides. The flexibility of its operation would allow UAVs to cover vast regions and traverse different types of ground with ease. During the mapping stage, the hybrid UAVs captured high-resolution imagery, from which farmers could obtain the information needed to assess the health of their crops by performing advanced spectral analysis. The data collected helped in identifying areas that required urgent attention, such as nutrient deficiency or pest infestation, and accurate mapping led to timely interventions that improved yield potential and efficiency in resource use. In addition, hybrid UAVs were observed performing spraying operations in the latter stage with high precision in the application of inputs over crop regions. Here, variable-rate technology was used, whereby the UAV determined the specific amount of input needed considering the health assessment that was conducted beforehand. This reduces waste and environmental impacts.

The long flight times allowed by fixed-wing designs, combined with the multi-rotor vertical takeoff efficiency, ensured that there was no need to cover large agricultural fields. The gathering of accurate and timely data before spraying operations reduced the use of chemicals and maximized application efficiency, thus promoting healthier crops with less environmental damage. Hybrid UAVs have shown versatility in addressing various tasks, such as comprehensive mapping and data collection, for efficient crop spraying within a mission and answering multiple operational needs. Hybrid designs reduce the number of UAVs required for applications, thus lowering operational costs, which makes the technology more affordable to farmers for the adoption of precision agriculture.

Financial solutions for small farms: The entry costs associated with UAV technology are substantially high, covering hardware-related factors such as drones and sensors, as well as software-related factors, including data processing tools. Multispectral or thermal sensors are normally required to effectively carry out precision agriculture. Such components greatly increase the total investment. A few studies claim that these technologies can recover several costs over time because efficiency is usually gained through crop monitoring, irrigation management, and a decrease in pesticide use. This cost–benefit relationship holds true even when accounting for increased yields and decreased input costs [31,32].

In the context of small farms, lease programs, government subsidies, and cooperative ownership models have become the most practical remedies for easing financial constraints. For example, most governments provide subsidies for precision agriculture technologies in the context of reducing the cost of purchasing a UAV. Even shared ownership models have proven to be quite productive in many regions, as there are many small farms coming together to pool their investments and share an individual UAV system [33].

4. Challenges, Solutions, Open Issues, and Future Research Directions

In this section, we analyze the challenges in deploying UAVs for precision agriculture, possible solutions, open research issues, and future research directions.

4.1. Challenges and Possible Solutions in Precision Agriculture

The integration of UAV systems with AI-driven perception algorithms has great potential in the field of precision agriculture. However, UAVs present a plethora of technical challenges when operating in agricultural environments, from flight stability and energy efficiency to issues associated with the adaptability of AI models to crop- and environment-based variations. Furthermore, the poor quality of data, delay in transmitting data, and the expense involved in high-resolution imaging and processing create more barriers. The removal of these barriers is also needed so that precision agriculture can realize all of its potential through AI-empowered UAV systems. The challenges and possible solutions for UAV-driven AI systems concerned with precision farming are listed in Table 8.

Table 8. AI-driven UAVs for precision agriculture: challenges and possible solutions.

Challenge	Details	Possible Solutions	Use Cases
Flight stability and reliability	Small UAVs struggle with stability in turbulent conditions, which impacts image quality and data accuracy.	Developing more robust airframes, improving flight control algorithms, and enhancing UAV navigation for better obstacle avoidance and disturbance rejection.	UAV applications in vineyards and orchards, where flight stability ensures high-quality data capture [65].
Payload and endurance	UAVs have limited payload capacity and short flight times due to low battery capacity and high power consumption.	Develop higher-energy-density batteries (e.g., lithium–sulfur), use hydrogen fuel cells or solar power, and reduce payload weight with miniaturized sensors. Alternatively, opt for larger UAV platforms.	UAVs using hydrogen cells (HyDrone series) demonstrate longer flight times, which are suitable for extensive monitoring [91].
Image resolution and cost	High-resolution imaging is expensive and requires trade-offs in flight efficiency.	Use hyperspectral and multispectral imaging to focus on critical wavelengths, optimize flight paths, and adopt edge computing for real-time data processing.	The high-resolution imaging of vineyards allows for efficient nutrient management [92].
Energy efficiency	UAVs run on limited battery life, and AI models are power hungry.	Developing lightweight AI models and using energy-efficient algorithms or edge computing to reduce power consumption during flight. Moreover, this introduces power-saving strategies like dynamic task allocation between edge and cloud computation.	In-field edge processing has been used in vineyards to extend UAV battery life and optimize flight paths [93].
Benchmark dataset availability	The lack of large-scale labeled datasets for training AI models slows development in precision agriculture applications.	Develop UAV-specific benchmark datasets with annotated data for various crop conditions, stress stages, and regions. Transfer learning on preexisting datasets can be useful for enhancing generalization.	Transfer learning applied in weed detection applications has helped reduce the dataset requirements [42].
Algorithm generality	AI models are not easily adaptable to new environments, crops, or conditions.	Implement transfer learning and domain adaptation techniques to improve the generalization of AI models across different agricultural environments.	Algorithm adaptation has been attempted in cross-crop settings to increase applicability [94].
Precision in data collection	Variability in UAV altitude and environmental conditions (e.g., sunlight) can cause inconsistent data collection from sensors like multispectral and hyperspectral cameras.	Implement adaptive UAV flight planning with real-time sensor calibration and incorporate machine learning algorithms to adjust for environmental changes during data acquisition.	Multispectral UAVs with adaptive planning are used for chlorophyll assessment under varying sunlight conditions [95].
Data transmission in remote areas	Limited network connectivity in rural or remote farming areas hampers real-time data transmission and UAV operation.	Use low-power wide-area networks (LPWANs) combined with edge computing for local processing and data transmission via satellites in critical scenarios.	LPWAN and edge computing have been applied to enable localized data processing for UAVs on rural farms [96].

4.2. Open Research Issues and Future Research Directions for Precision Agriculture

The integration of AI algorithms and sensor technologies with UAVs has significant potential for revolutionizing precision farming. This convergence aims to improve agricultural efficiency by enabling more accurate data collection, real-time analysis, and autonomous decision making. Future research and development in this domain are likely to focus on the following areas.

4.2.1. Enhanced AI Algorithms for Adaptive Learning

Future AI models will emphasize adaptive learning algorithms [97–99] capable of adjusting to varying environmental conditions, crop types, and geographical regions. Traditional supervised learning algorithms, such as SVMs and random forests, perform well but are limited by their dependency on large labeled datasets. New directions involve deploying reinforcement learning and generative adversarial networks (GANs) for real-time adaptation to environmental variations and unseen data distributions [100]. For example, adaptive AI can help UAVs autonomously adjust flight paths based on real-time wind speed data or changing crop health conditions, thereby reducing the need for constant human intervention.

4.2.2. Data Fusion from Multi-Sensor Networks

UAVs are equipped with various sensors (e.g., RGB, multispectral, hyperspectral, and LIDAR) that provide a wide range of data. Integrating multi-sensor fusion algorithms will allow for more comprehensive analyses, particularly when combining surface and subsurface data [101,102]. Future research will explore data fusion algorithms such as Kalman filters, Bayesian networks, and deep-learning-based fusion frameworks to seamlessly combine aerial and ground-based sensor data. For example, fusing UAV data with soil moisture sensors can provide more accurate predictions for irrigation management and crop stress detection.

4.2.3. Lightweight and Energy-Efficient AI Models

With constraints on UAV battery life and available computational power, future approaches should be focused on developing lightweight AI models that are also computationally and energy efficient [103]. Edge AI and federated learning will therefore play an important role in enabling AI models to run locally on UAVs and reduce dependence on heavy data transmission to remote servers [104]. Pruned neural networks and quantized models may achieve very high accuracy with minimal computational load, thus saving energy during UAV operation. For example, energy-efficient AI for UAVs can increase coverage in the field, collecting data over longer periods without the need for frequent recharging.

4.2.4. Improved 3D Mapping and Reconstruction

Advances in computer vision algorithms will play a crucial role in improving the UAV-based 3D reconstruction of agricultural fields. Simultaneous localization and mapping (SLAM) algorithms will continue to evolve to produce highly accurate, real-time 3D field and crop models that will streamline the tracking of crop height, biomass, and health, among others [105]. For example, advanced SLAM algorithms on UAVs can monitor dynamic changes in plant growth and hence provide immediate warnings relating to the potential occurrence of diseases or nutrient deficiency issues much earlier than usual.

4.2.5. Scalability with 5G and Beyond

The 5G network will increase the scalability of UAV systems in precision agriculture, as it provides high bandwidth for low latency and immediate data transmission [106]. This will enable UAV fleets to cover larger fields and send data back to centralized systems without significant delays [107]. For example, with 5G, UAVs can transmit high-resolution

crop health data in real time to allow for proper adjustments to be made immediately away from irrigation or fertilization strategies.

4.2.6. Multi-UAV Formation and Automation

Researchers have also explored swarm control techniques that employ multiple UAVs to conduct a variety of tasks effectively. The use of swarming technology can significantly reduce battery costs, increase efficiency, and shorten lighting time. In agriculture, UAVs are used to apply fertilizers and pesticides to croplands, although all of these tasks are still performed manually. UAVs in agriculture must be automated to perform active tasks, such as monitoring crop health, spraying, and applying fertilizers and pesticides in crop fields. This requires a complex combination of developments in UAV swarming, flight control, stability, aerodynamics, and computer vision.

4.2.7. User-Friendly Smartphone Applications

Finally, it is necessary to develop user-friendly smartphone and browser-based applications to handle UAV and sensor applications. UAV operation is typically difficult for individuals working in agriculture. A more intuitive user interface can make UAV control easier for older or unfamiliar users. By reducing the dependency on specialized operators for field monitoring, UAV technologies can be adopted by farmers more readily. In particular, human-centered user interfaces and feedback are effective for dealing with multi-UAV systems.

4.2.8. Need for Benchmark Datasets in Agricultural AI

The primary difficulty in developing agricultural AI is the scarcity of large, labeled, and domain-specific benchmark datasets. This severely limits the training and validation of AI models for generalization to other crop conditions, phenological stages, and environmental conditions. In future studies, UAV-specific benchmark datasets should be created across a range of crop types, stress indicators, and environments. Such datasets would enhance the robustness of the models and enable the scaling up of AI solutions for multiple applications in agriculture [42]. More generic datasets, such as ImageNet, have been used earlier to pretrain models in agriculture, but they do not contain specific agricultural content that fails to capture finer nuances in crop health or region-specific phenotypic differences [18]. To improve the predictive accuracy of UAV-based monitoring, research on agricultural AI must shift toward specifically dedicated benchmarks.

4.2.9. Standardized Testing Environments

Standardized testing environments are urgently needed to guarantee the consistent evaluation and reproducibility of AI models in agricultural UAV applications. The absence of common testing protocols currently hampers comparability across studies, limiting the progress of the field in developing robust, field-ready UAV-AI solutions. Standardized frameworks can be based on both real-world and simulated conditions while considering the following key variables: environmental diversity, including weather and lighting, sensor types, and crop diversity. These are all supportive of the cross-validation and alignment of methodologies [108].

4.2.10. Computational Power

UAVs are typically power constrained, particularly when they carry power-hungry sensors or fly over complex environments that require significant processing. The need for power constraints on UAVs implies that the AI models deployed on them must be efficient and low power. The techniques that were researched included model pruning, which removes redundant weights in neural networks; quantization, where lower precision weights are used; and distillation, training smaller models using knowledge from larger ones. These are especially helpful for edge computing settings, such as UAVs, which are characterized by limited computing power and energy.

5. Conclusions

The integration of UAVs with AI, ML, and advanced sensor technologies presents significant opportunities to revolutionize precision agriculture by improving efficiency, accuracy, and sustainability. UAVs equipped with high-resolution multispectral, hyperspectral, and thermal sensors provide critical data on crop health, soil conditions, and pest infestation in real time. Data gathered through such sensors and analyzed through AI and ML models enable the system to make specific decisions to optimize resource usage and achieve better crop management. The benefits include labor cost savings, increased yield, and relief of the environment from water, fertilizers, and pesticide usage. However, problems continue with operating costs, short-lived batteries, and demands on the environment. Another problem that has been a significant issue is that linking very high-technology sensors with UAVs raises issues about cost-effectiveness and user-friendliness. This requires the further miniaturization of sensors, more energy-efficient AI models, and fully autonomous UAV systems, all of which are predicted to fuel the next wave of innovation in precision agriculture. In this paper, we present a brief overview of AI-driven UAV technologies for precision farming. Various AI and ML algorithms have been thoroughly reviewed and analyzed based on their capabilities and potential utility. We provided an overall technical view of the current research progress on AI-driven UAV technologies in precision farming. Furthermore, we have provided a list of open issues and future research challenges at the end of the article.

Future trends in general areas of application will include the enhanced efficiency of farming processes through automation involving AI-empowered UAV systems. AI-empowered UAV systems will be able to monitor and intervene in real time with minimal human intervention. Nevertheless, an even more robust, adaptable sensor system could be developed with a focus on higher data accuracy and applicability in different agricultural environments. Ensuring the privacy and security of data while minimizing the socioeconomic implications of such automation on agriculture is critical for employing these technologies effectively. The future of more resistance, efficiency, and sustainability in agriculture will produce resilience for agricultural applications through UAV-AI-sensor integration, enabling a world food security outcome for mankind against increasing climate change.

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